

BEE 4750 Lab 2: Monte Carlo

Name: Regina Velasco

ID: rv297

Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()

Pkg.add("Distributions")
Pkg.add("DataFrames")
Pkg.add("StatsBase")
Pkg.add("Plots")
```

```
Activating project at `~/BEE 4750 /Lab 2/lab-2-reginavelasco`
  Resolving package versions...
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Project.toml`
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Manifest.toml`
  Resolving package versions...
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Project.toml`
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Manifest.toml`
  Resolving package versions...
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Project.toml`
```

```
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Manifest.toml`  
Resolving package versions...  
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Project.toml`  
No Changes to `~/BEE 4750 /Lab 2/lab-2-reginavelasco/Manifest.toml`
```

```
using Random # random number generation  
using Distributions # probability distributions and interface  
using Statistics # basic statistical functions, including mean  
using Plots # plotting  
import Pkg
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

`TaskLocalRNG()`

It doesn’t matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don’t re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you’re working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#   is the mean and   is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for ar
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

100-element Vector{Int64}:

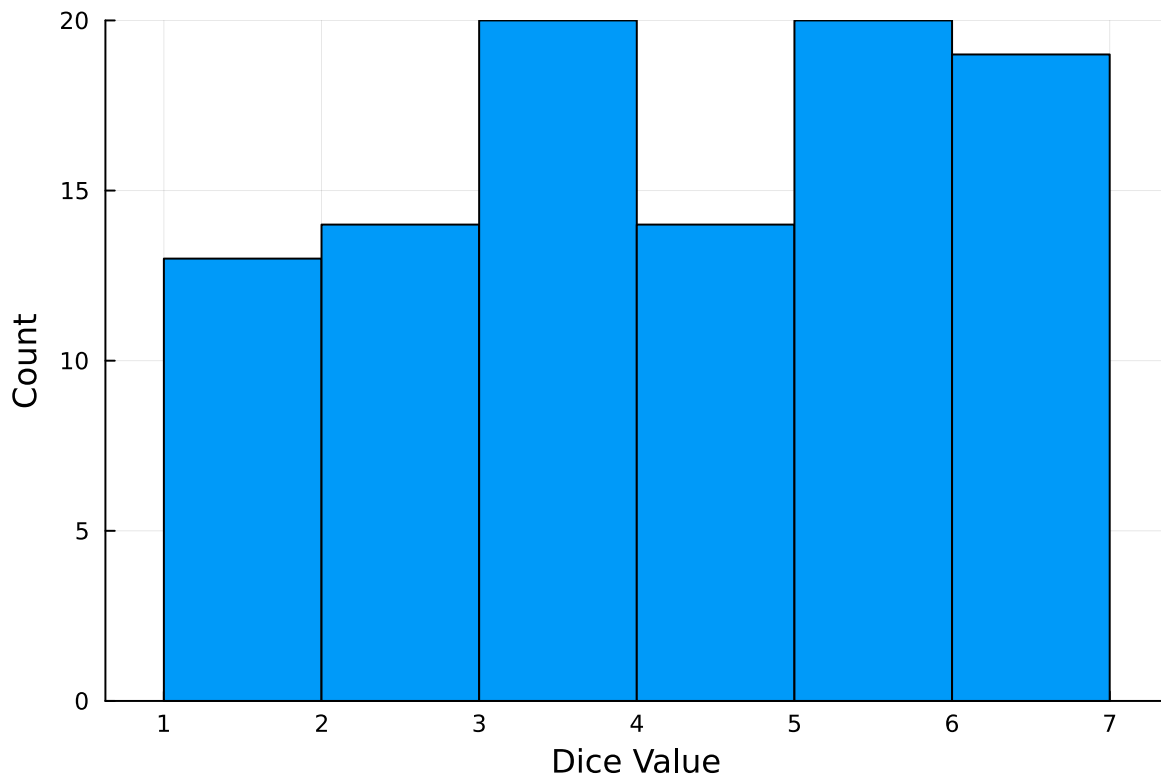
```
1
3
5
4
6
2
5
5
5
2

3
6
5
5
6
3
```

6
6
6

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)  
ylabel!("Count")  
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a Run All command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
using Distributions, DataFrames, Random, StatsBase, Plots

L = 200000 #max loss, USD
k = 0.8 #slope
h0 = 3.0 #inflection point, m

Random.seed!(1)

#depth-damage function

function damage(h)
```

```

    if h > 0
        d = L / (1 + exp(-k * (h - h0)))
    else
        d = 0
    end
    return d
end

N = 500000

# Model 1: LogNormal(=1.2, =0.3)
ln_dist = LogNormal(1.2, 0.3)
ln_samples = rand(ln_dist, N)
ln_dmg = [damage(h) for h in ln_samples]

# Model 2: GeneralizedExtremeValue(=3, =0.9, =-0.15)
gev_dist = GeneralizedExtremeValue(3, 0.9, -0.15)
gev_samples = rand(gev_dist, N)
gev_dmg = [damage(h) for h in gev_samples]

# histograms
h1 = histogram(ln_dmg, legend=false, xlabel="Damages (USD)", ylabel = "Frequency", title="L")
display(h1)

h2 = histogram(gev_dmg, legend=false, xlabel="Damages (USD)", ylabel = "Frequency", title="G")
display(h2)
"""

batches = 50

function batch_stats(x; q=0.99, B=50)
    n = length(x) ÷ B
    means = [mean(x[(b-1)*n+1:b*n]) for b in 1:B]
    qs = [quantile(x[(b-1)*n+1:b*n], q) for b in 1:B]
    mean_full = mean(x)
    q_full = quantile(x, q)
    se_mean = std(means) / sqrt(B)
    se_q = std(qs) / sqrt(B)
    return mean_full, se_mean, q_full, se_q
end

ln_mean, ln_mean_se, ln_q99, ln_q99_se = batch_stats(ln_dmg)
gev_mean, gev_mean_se, gev_q99, gev_q99_se = batch_stats(gev_dmg)

```

```

println("LogNormal mean = \\\$(round(ln_mean, digits=2))")
println("LogNormal 99th percentile = \\\$(round(ln_q99, digits=2))")
println("GEV mean = \\\$(round(gev_mean, digits=2))")
println("GEV 99th percentile = \\\$(round(gev_q99, digits=2))")

"""

# sample size
N = length(ln_dmg)

# standard deviation of simulated damages
ln_std = std(ln_dmg)
gev_std = std(gev_dmg)

# Monte Carlo standard error using  $1/\sqrt{N}$ 
ln_se = ln_std / sqrt(N)
gev_se = gev_std / sqrt(N)

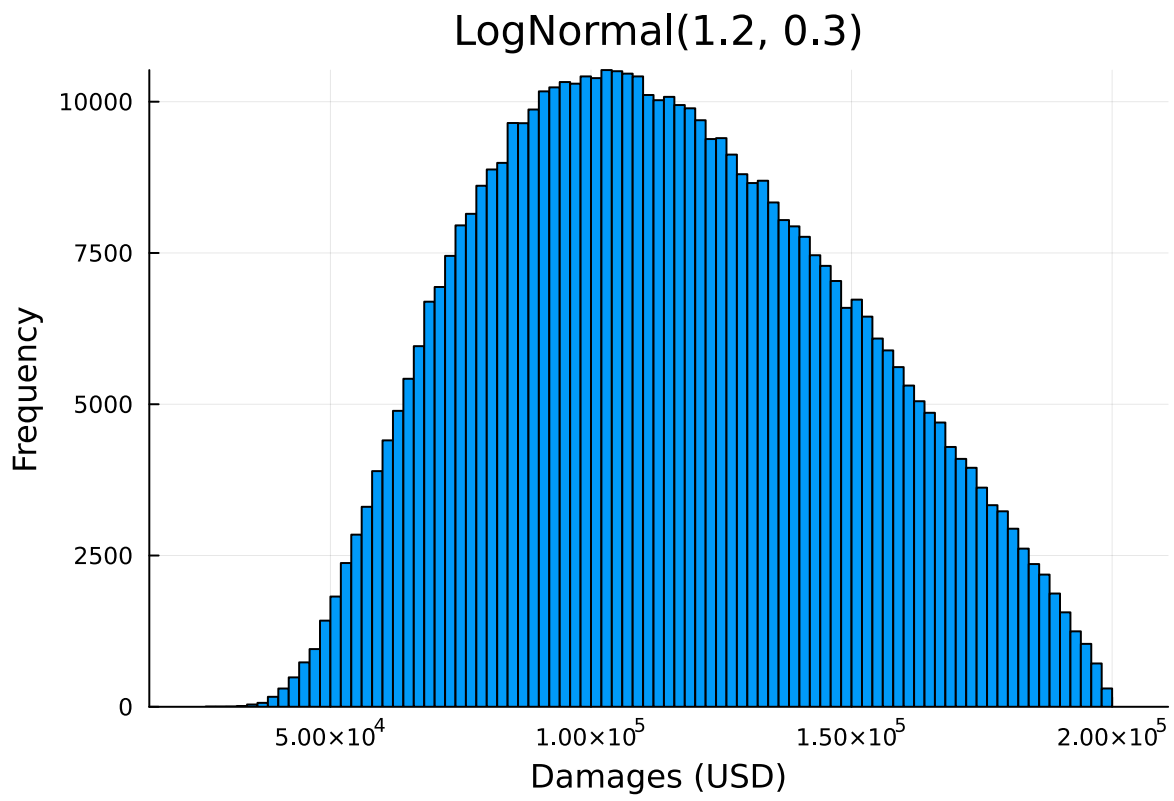
# sample means
ln_mean = mean(ln_dmg)
gev_mean = mean(gev_dmg)

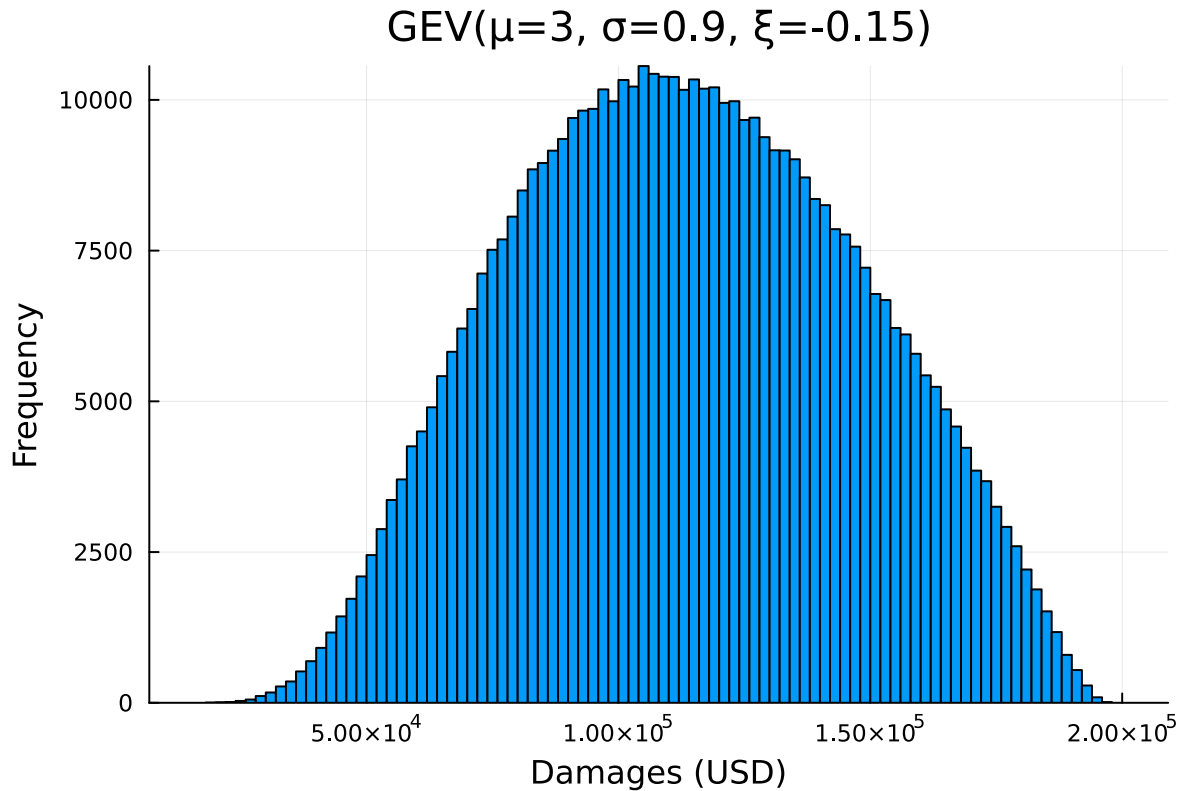
# 99th percentile
ln_q99 = quantile(ln_dmg, 0.99)
gev_q99 = quantile(gev_dmg, 0.99)

# 95% confidence intervals using CLT
z = 1.96
ln_ci = (ln_mean - z*ln_se, ln_mean + z*ln_se)
gev_ci = (gev_mean - z*gev_se, gev_mean + z*gev_se)

println("LogNormal mean damage = \\\$(round(ln_mean, digits=2)) ± \\\$(round(ln_se, digits=2))")
println("95% CI for LogNormal = [\\$(round(ln_ci[1], digits=2)), \\$(round(ln_ci[2], digits=2))")
println("GEV mean damage = \\\$(round(gev_mean, digits=2)) ± \\\$(round(gev_se, digits=2))")
println("95% CI for GEV = [\\$(round(gev_ci[1], digits=2)), \\$(round(gev_ci[2], digits=2))")
println()
println("LogNormal 99th percentile = \\\$(round(ln_q99, digits=2))")
println("GEV 99th percentile = \\\$(round(gev_q99, digits=2))")

```





LogNormal mean damage = \$115198.5 $\pm$ \$48.42
 95% CI for LogNormal = [\$115103.59, \$115293.4]
 GEV mean damage = \$113466.03 $\pm$ \$47.63
 95% CI for GEV = [\$113372.67, \$113559.4]

LogNormal 99th percentile = \$189851.32
 GEV 99th percentile = \$183352.17

- **What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?**

For the LogNormal distribution, the Monte Carlo estimate of the expected value is \$115198.5 and the 99% quantile is \$189851.32. For the GEV distribution, the Monte Carlo estimate of the expected value is \$113466.03 and the 99% quantile is \$183352.17.

- **** How did you decide on your sample size? ****

We saw in class that decreasing Monte Carlo error by 10x requires 100x additional samples. With sample size $N = 500,000$ the Monte Carlo standard error, $SE = \frac{\sigma}{\sqrt{N}}$, was less than \$200 for both distributions. That is less than 0.2% of the mean, meaning

this is already a pretty good model. In addition, the 95% confidence intervals for both methods were very small too (less than 0.2% of the mean). Therefore, increasing the N would not really change the estimated mean by much and this is how I chose 500,000 as a solid sample size to yield good results.

- ** Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)? ** The models were pretty similar. They both neared maximum damages of \$200,000 and are bell-shaped as shown above in the graphs. However, the LogNormal model has a little bit of a higher mean and 99th percentile damage than the GEV model. This makes sense because the graph shows that the LogNormal graph has a bit of a tail on the right-side. This means that it had a higher number of severe floods above the inflection point h_0 .
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

References

Put any consulted sources here, including classmates you worked with/who helped you.